# Robed Beauvile-Owen

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## EXPERIENCE

**Founder & Data Scientist** — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy delivering site-selection analyses for retail and service clients, including a $1.5M expansion opportunity.
- Engineered Python ETL pipelines automating geospatial feature ingestion and outcome-distribution reporting, cutting market analysis turnaround by 91% (weeks to days).
- Built a Monte Carlo risk framework modeling correlated stochastic inputs (demand, cost, competition), used by clients to stress-test expansion decisions before committing capital.
- Designed a proprietary site-scoring framework integrating U.S. Census demographic data, traffic patterns, competitive density, consumer behavior, and geospatial analytics to estimate revenue potential and location risk.

**Senior Data Scientist** — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap in partnership with strategy, engineering, and behavioral science leadership, establishing OKRs and strategic analytics initiatives.
- Applied predictive modeling, customer segmentation, and uplift modeling to identify high-value Medicare/Medicaid eligible cohorts, reducing ad spend by $10K per campaign across a pipeline of millions of healthcare profiles.
- Designed, deployed, and standardized cross-team experimentation and causal inference frameworks for direct-mail campaign optimization, applying CUPED to reduce metric variance by 20%, enabling smaller samples and faster test reads.
- Re-architected an end-to-end distributed churn prediction platform using Python, SQL, Snowflake, Spark, and Databricks, reducing runtime by 48% and memory consumption by 23% via pipeline code optimization.

**Machine Learning Engineer** — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- Defined quantitative success metrics and established performance evaluation frameworks for production search and regression models using offline and online experimentation.
- Developed a regression-based shipping price API that accurately breaks down bulk order history to estimate individual item shipping costs; scaled the system to handle 10,000+ daily requests while maintaining strict performance standards.
- Partnered with product and engineering teams to reduce search customer dissatisfaction by 12% via custom Elasticsearch indexing, tokenizers, and TF-IDF weighting, lifting user engagement.

**Data Scientist** — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Developed and evaluated XGBoost and other classification models using millions of healthcare consumer records and hundreds of appended external attributes, driving a 3% lift in response rates.
- Streamlined reporting and visualization analytics pipelines via Python, SQL, Snowflake, and Power BI; removed 60 minutes of manual daily overhead while strengthening data scalability.
- Improved geospatial targeting accuracy by 20% through GIS analysis with GeoPandas and spatial modeling, enabling precise opportunity identification and resource allocation.

**Graduate Researcher & Teaching Assistant** — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Conducted original doctoral research spanning mathematical optimization, functional analysis, and operator theory, producing 3 substantial results addressing complex theoretical problems to complete doctoral dissertation requirements.
- Taught undergraduate mathematics and statistics courses (75-200 students) and coordinated summer programs for 100+ students, achieving a 95% pass rate.

## TECHNICAL PROJECTS

**Quantitative Economics Platform** github.com/beauvilerobed/quantitative-economics-with-python

**Technologies:** Jupyter, NumPy, SciPy

- Built an open-source research platform modeling stochastic dynamics with Markov chains, survival analysis, and Kalman filtering, the state-transition methods behind customer retention and churn forecasting.

**Customer LTV & Churn Survival Modeling** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

**Technologies:** Python, NumPy, Survival Analysis, Monte Carlo Simulation

- Simulated 1M customer lifetimes under constant and tenure-dependent churn hazards; showed the LTV = ARPU / churn shortcut fails when churn varies with tenure and estimated customer lifetime value from empirical survival curves.

**CaLaun** — *AI-Powered Calculus Tutor Platform* github.com/calaun-learning/calaun

**Technologies:** Python, LLM Integration, Django, PostgreSQL

- Designed an open-source calculus tutoring platform that combines symbolic mathematics with large language models to generate accurate, step-by-step calculus solutions, serving 400+ weekly users with 90+ automated tests to secure deployment reliability.

## EDUCATION

**Ph.D. in Mathematics** — *University of Pittsburgh* 2017 - Dec 2023

**Dissertation:** A Minimal Invariant Set, Frames, and Operators

**B.S. in Mathematics** — *Florida State University* 2014 - 2017

**Minor:** Philosophy | **GPA:** 3.90 | **Undergraduate Thesis:** Homotopy Type Theory, Univalent Foundations, and Binary Trees

## TECHNICAL SKILLS

**Programming:** Python, SQL, R, C#, Bash, PowerShell

**Statistics & Experimentation:** Causal Inference, A/B Testing, CUPED, Power Analysis, Hypothesis Testing, Difference-in-Differences, Synthetic Controls, Conditional Average Treatment Effects (CATE)

**Machine Learning:** XGBoost, Scikit-learn, Regression, Classification, Clustering, Uplift Modeling, Monte Carlo Methods, Stochastic Modeling, LLMs (Llama)

**Data Engineering & Cloud:** Spark, Databricks, Snowflake, Azure, SQL Server, PostgreSQL, Docker, Elasticsearch, ETL/ELT Pipelines, Feature Engineering

**Data Visualization & Analytics:** Power BI, Tableau, Excel (Pivot Tables), Matplotlib, Seaborn, Plotly, Executive Reporting, Predictive Analytics

**Leadership:** Technical Roadmaps, Stakeholder Management, Cross-Functional Collaboration, OKRs/KPIs, Executive Communication